

Séminaire EDL

Sacha Braun & Guillaume Principato



Need for uncertainty quantification

- Predicting EV charging session for smart charging

The importance of understanding EV charging behaviours



RTE: Integration of electric vehicles into the power system in France

Need for uncertainty quantification

- Predicting EnR for trading on the energy market



The banner features a split background: the left side shows a vast field of offshore wind turbines on a blue sea under a cloudy sky, while the right side shows a close-up of solar panels reflecting a bright sunset. The text is overlaid on this background.

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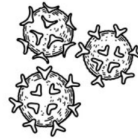
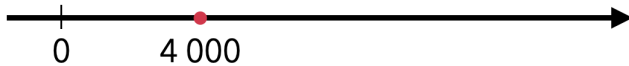
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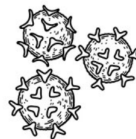
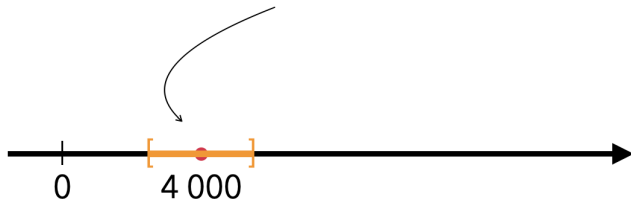
Need for uncertainty quantification



White blood cells
per microliter for
one individual

Need for uncertainty quantification

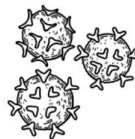
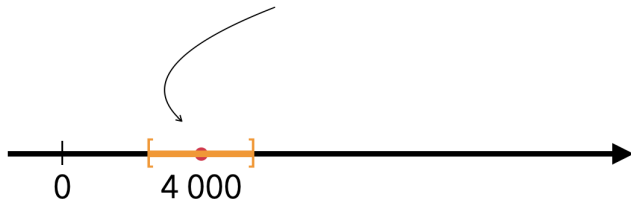
$$\mathbb{P}(\text{White blood cells} \in IC) = 1 - \alpha$$



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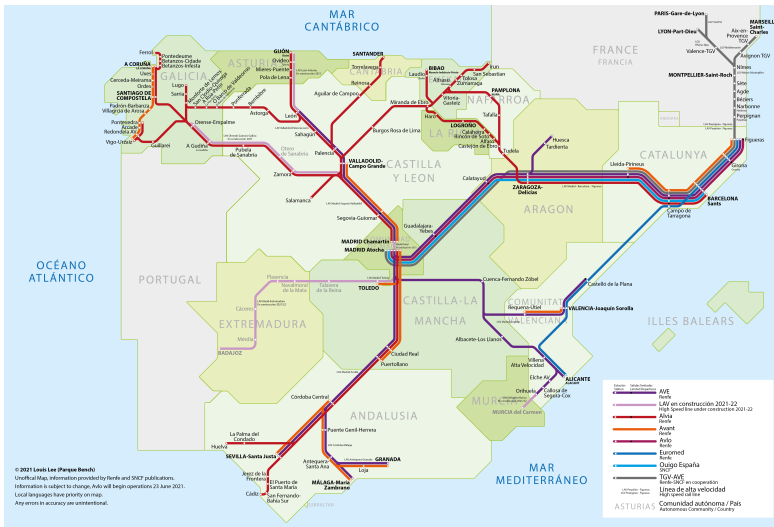
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Difficulty: High-dimensional inputs/outputs, no distributional assumptions

Need for uncertainty quantification

- Predicting delays on train networks



Need for uncertainty quantification

- Reinforcement learning and robotics



Table of Contents

1. Foundations of conformal prediction
2. Evaluating conditional miscoverage
3. Conformal Prediction for Hierarchical Data
4. Gaussian conformal prediction
 - 4.1 Partially Revealed Outputs
 - 4.2 Projection of the output
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Conformal prediction yields finite sample guarantees

- **Observation:** dataset $\mathcal{D}$ of n independent and identically distributed (i.i.d.) samples $(X_i, Y_i) \sim \mathbb{P}_{X,Y}$ with $(X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}$ for $\mathcal{Y} \subseteq \mathbb{R}$
- **Goal:** Given a target miscoverage level $\alpha \in (0, 1)$, construct a **set-valued predictor** $C_\alpha(X)$ such that

$$\mathbb{P}_{X,Y}(Y_{n+1} \in C_\alpha(X_{n+1})) \approx 1 - \alpha$$

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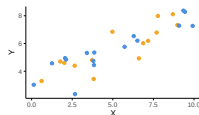
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- **Conformal prediction allows **finite sample guarantees****
(Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023; Angelopoulos et al., 2024)
- **Alternative frameworks:** Bayesian, resampling, quantile regression (QGAM), etc.

A practical example: SCP with absolute residuals

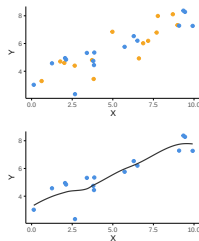
Split the data between $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$



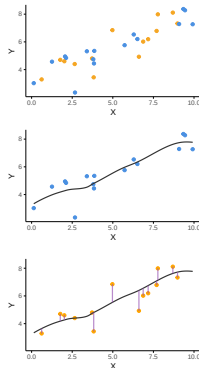
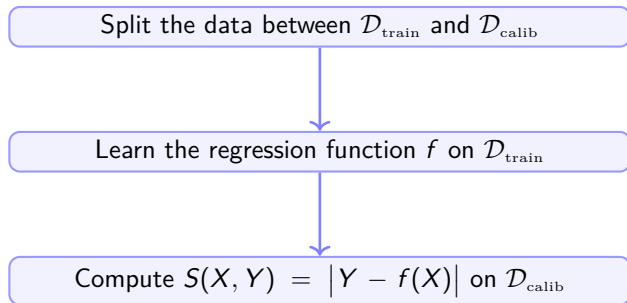
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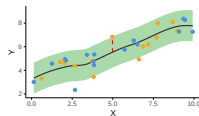
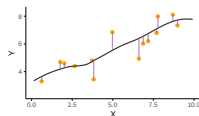
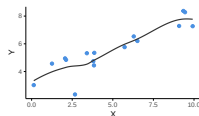
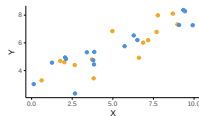
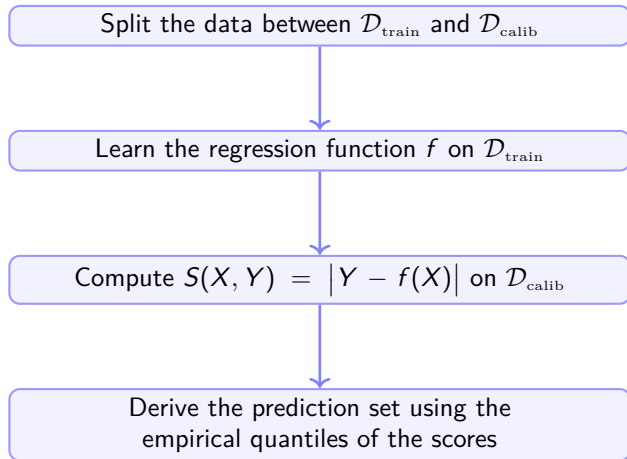
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We get marginal guarantees and no conditional ones

Given fixed symmetric “nonconformity score function” $S : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, we get

- $\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \approx 1 - \alpha$ ✓

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Impossible to get conditional validity without additional assumptions
(see, e.g., [Vovk 2012](#); [Foygel Barber et al. 2021](#); [Lei and Wasserman 2014](#))

How to get good marginal guarantees in practice

Good conditional coverage $\iff$ score functions that “capture uncertainty”
and have weak dependence on X

Choice of score function is crucial

- Any (even estimated) score can be “conformalized” using SCP
- Scores can be based either on mean estimates or quantile estimates
- Only good scores yield meaningful prediction sets

Improved conditional coverage with standardized residuals

- Learn **any** model f on the training dataset **AND** learn $\sigma(X)$, to predict the standard deviation of $|Y - f(X)|$
- Define the non-conformity score ([Lei et al., 2018](#))

$$S(X, Y) = \frac{|Y - f(X)|}{\sigma(X)}$$

- Let $\hat{q}_\alpha$ be the empirical quantile of the calibration scores of level $1 - \alpha$

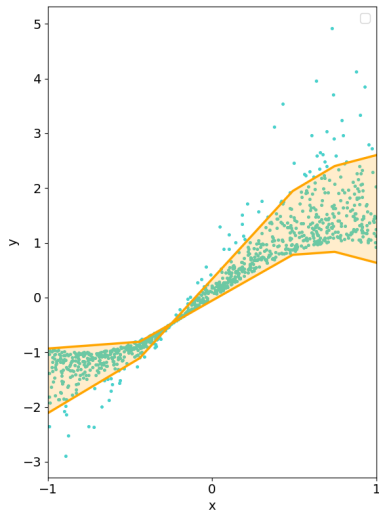
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- Output the prediction set

$$C_\alpha(X) = \{y \in \mathcal{Y} : f(X) - \sigma(X)\hat{q}_\alpha \leq y \leq f(X) + \sigma(X)\hat{q}_\alpha\}$$



Conformalized Quantile Regression

- Learn quantile estimates $\hat{y}_{\alpha/2}(X)$ and $\hat{y}_{1-\alpha/2}(X)$ on the training set. Typically by minimizing the **pinball loss**, for $\tau = \alpha/2$ and $\tau = 1 - \alpha/2$:

$$L_{\tau}(y, \hat{y}_{\tau}) = \begin{cases} \tau(y - \hat{y}_{\tau}) & \text{if } y \geq \hat{y}_{\tau}, \\ (1 - \tau)(\hat{y}_{\tau} - y) & \text{if } y < \hat{y}_{\tau}. \end{cases}$$

- Define the score function ([Romano et al., 2019](#))

$$S(X, Y) = \max\{\hat{y}_{\alpha/2}(X) - Y, Y - \hat{y}_{1-\alpha/2}(X)\}$$

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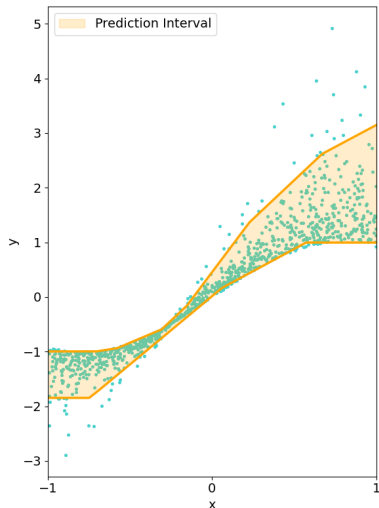
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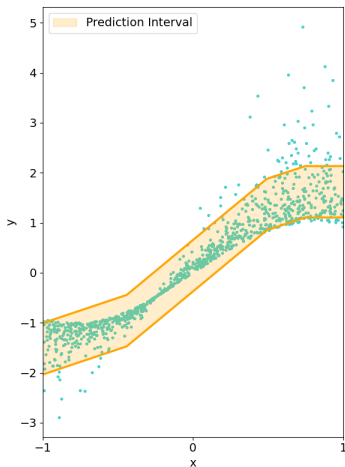
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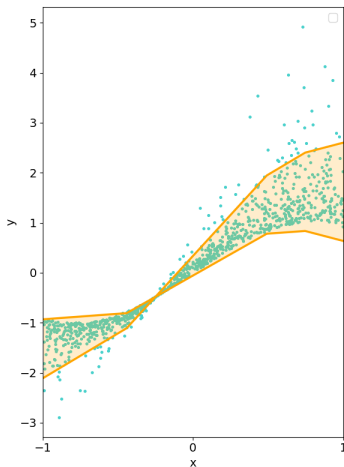
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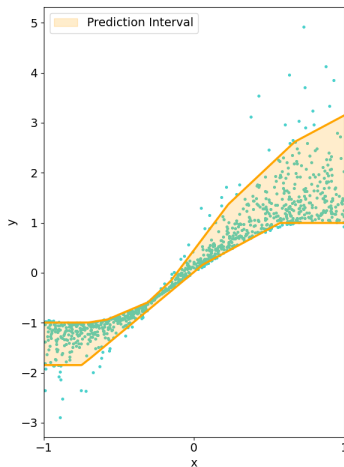
Conditional validity for these three score functions



(a) Absolute residuals



(b) Standardized residuals



(c) CQR

Three open research axes in Conformal Prediction

We focus on three active and open directions in conformal prediction:

- **Conditional coverage**
 - Conditional Coverage Diagnostics for Conformal Prediction, [Braun et al. \(2025c\)](#)

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- **Multivariate extensions**

- Conformal Prediction for Hierarchical Data, [Principato et al. \(2024\)](#)
- Minimum volume conformal sets for multivariate regression, [Braun et al. \(2025a\)](#)
- Multivariate conformal prediction via conformalized gaussian scoring, [Braun et al. \(2025b\)](#)

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 - Multivariate conformal prediction via conformalized gaussian scoring, [Braun et al. \(2025b\)](#)
- **Beyond the i.i.d. setting**
 - Adaptive Conformal Inference through the lens of Blackwell Approachability [Principato and Stoltz \(2025\)](#)

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Excess risk of the target coverage

- For any data-dependent set C_α , let $Z = \mathbb{1}\{Y \in C_\alpha(X)\} \in \{0, 1\}$
- Perfect conditional coverage if $p(X) := \mathbb{P}(Y \in C_\alpha(X)|X) = \mathbb{P}(Z = 1|X) = 1 - \alpha$ a.s.

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- **Key idea:** Estimating $\mathbb{P}(Z = 1|X = x)$ is a binary classification problem
 - Classification task equivalent to minimizing the risk

$$\mathcal{R}_{\ell_2}(h) = \mathbb{E}[(h(X) - Z)^2]$$

- Minimizer of $\mathcal{R}_{\ell_2}$ satisfies $h^*(X) \in \mathbb{E}[Z | X] = p(X)$
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- **“Excess risk of target coverage”** ([Braun et al., 2025c](#))

$$\ell\text{-ERT} = \mathcal{R}_\ell(1 - \alpha) - \mathcal{R}_\ell(p) = \mathcal{R}_\ell(1 - \alpha) - \inf_h \mathcal{R}_\ell(h)$$

- Provides a measure of miscoverage and a detailed diagnostic of over/under coverage
- Can be estimated with standard tools from non-parametric estimation

Benefits: a set of interpretable metrics

- Property of excess risk for a proper loss

$$\ell\text{-ERT} = \mathcal{R}_\ell(1 - \alpha) - \mathcal{R}_\ell(p) = \mathbb{E}_X[d_\ell(1 - \alpha, p(X))]$$

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- Examples of proper scoring rules and their associated ERT scores

Name	Proper score $\ell(p, y)$	ℓ -ERT
L_1 -ERT	$\text{sgn}(p - (1 - \alpha))(1 - \alpha - y)$	$\mathbb{E}_X[1 - \alpha - p(X)]$
L_2 -ERT (Brier score)	$(y - p)^2$	$\mathbb{E}_X[(1 - \alpha - p(X))^2]$
KL-ERT (Log loss)	$-\log p_y$	$\mathbb{E}_X[D_{\text{KL}}(p(X) 1 - \alpha)]$

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 - $\ell\text{-ERT}(h) > 0 \Leftrightarrow$ the strategy is not conditional.
- The empirical estimate is (where predictions must be out of sample with cross validation for instance)

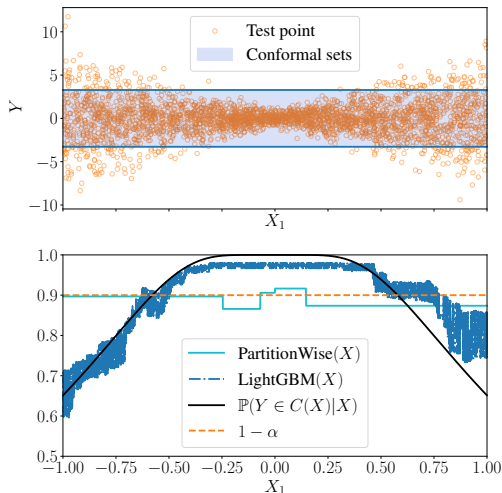
$$\widehat{\ell\text{-ERT}}(h) = \frac{1}{m} \sum_{i=1}^m [\ell(1 - \alpha, Z_i) - \ell(h(X_i), Z_i)],$$

- This is precisely what existing metrics such as CovGap are estimating but using partition wise predictors!

$$L_1\text{-ERT}(\text{PartitionWise}) \approx \text{CovGap}.$$

Metrics comparison without conditional coverage

- $X \sim \mathcal{U}([-1, 1]^8)$
- $Y \sim \mathcal{N}(0, \sigma(X_1))$
- Sets are obtained by doing conformal prediction with the score $S(X, Y) = |Y|$
- $L_1\text{-ERT}(\text{LightGBM}) = 0.09$
(boosting with decision trees)
- $L_1\text{-ERT}(\text{PartitionWise}) = 0.01$
(a classical diagnostic measure)



General diagnostics metrics comparison

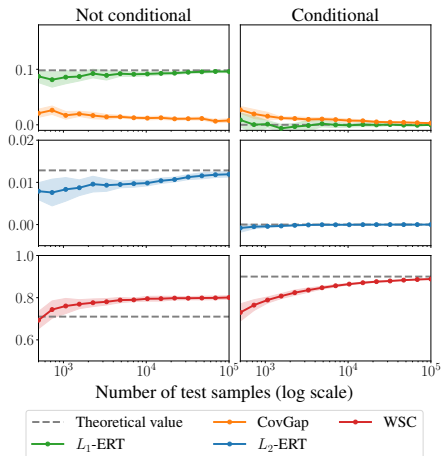


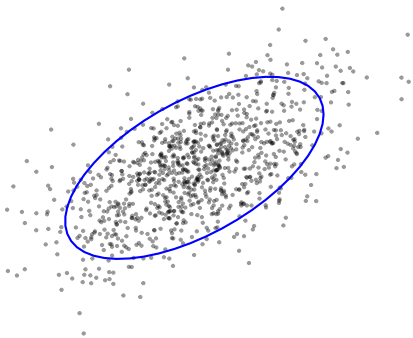
Figure: Estimated metrics as a number of test sample points. Top: WSC. Middle: CovGap and L_1 -ERT. Bottom axis: L_2 -ERT. **Left:** Standard CP not conditional. **Right:** Oracle conditional sets. Theoretical values are estimated using the true value $\mathbb{P}(Y \in C_\alpha(X) \mid X)$ with 300,000 samples from $\mathbb{P}_X$.

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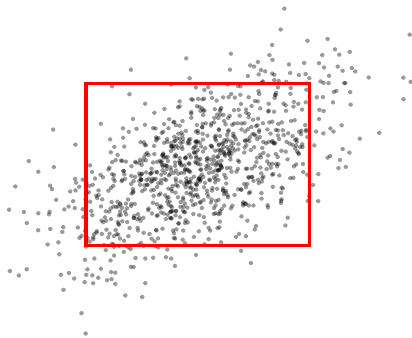
Prediction Regions for Multivariate Data: Validity

In regression for multivariate targets $\mathbf{y} \in \mathbb{R}^m$ and covariates $\mathbf{x} \in \mathbb{R}^d$ with $m > 1$, we can consider 2 types of prediction regions:



which targets joint coverage:

$$\mathbb{P}(\mathbf{y}_{T+1} \in E(\mathbf{x}_{T+1})) \approx 1 - \alpha$$



which targets component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in C_i(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

Ellipsoidal SCP (joint-coverage)

Challenge: build a real-valued score function that takes dependencies into account.

A flexible method consists in ellipsoidal conformal sets ([Johnstone and Cox, 2021](#)).

$$\check{s}_t = \|\mathbf{y}_t - \hat{\mathbf{y}}_t\|_A \stackrel{\text{def}}{=} \sqrt{(\mathbf{y}_t - \hat{\mathbf{y}}_t)^\top A (\mathbf{y}_t - \hat{\mathbf{y}}_t)},$$

with $\hat{\mathbf{y}}_t = \hat{\boldsymbol{\mu}}(\mathbf{x}_t)$ and matrix $A \succeq 0$.

Remark: if $A = \Sigma^{-1}$, the score is the *Mahalanobis* distance.

More about joint-coverage in Sacha's next part!

Component-wise SCP

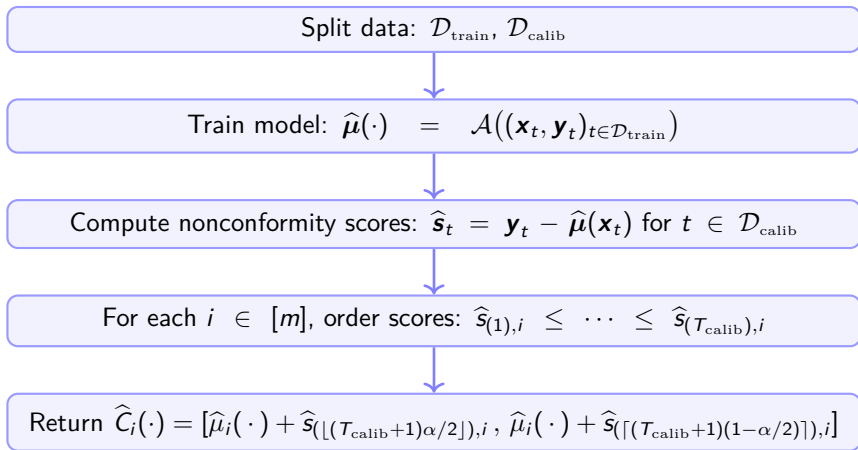
Underexplored objective, which appears naturally in industrial applications—with vectorial target under individual scrutiny.

Example: Load consumption on the twelve french regions estimated jointly, for which uncertainty should be component by component.

This leads to the natural component-wise coverage objective:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in C_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

Component-wise SCP: (baseline component-wise)



Immediate Results for the baselines (informal)

If the $(\mathbf{x}_t, \mathbf{y}_t)_{t \in [T+1]}$ are i.i.d., the $\hat{C}_i(\mathbf{x}_{T+1})$ from the Component-wise SCP algorithm achieve component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(\mathbf{y}_{T+1,i} \in \hat{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

Our Contributions

If $\mathbf{y}_{T+1,i}$ follow some hierarchical structure, adding a projection step improves efficiency.

Hierarchical Data

Denote $y_{B_1}, \dots, y_{B_n}$ the n first components of $\mathbf{y} \in \mathbb{R}^m$. The target $\mathbf{y}$ is hierarchical if

$$\mathbf{y} = H\mathbf{y}_{B_{1:n}} \quad \text{with for example} \quad H = \begin{bmatrix} \text{Id}_2 \\ 1 & 1 \end{bmatrix} \quad \text{which corresponds to} \quad y_{\text{Tot}} = y_{B_1} + y_{B_2}.$$

Example: Consider $\mathbf{y}_{B_{1:12}}$, the load consumption within the twelve french regions and y_{Tot} the french one. Then, $\mathbf{y}$ is hierarchical since $y_{\text{Tot}} = y_{B_1} + \dots + y_{B_{12}}$.

Coherent Forecast

Problem: In general, $\hat{\mathbf{y}} = \hat{\boldsymbol{\mu}}(\mathbf{x})$ does not satisfy the linear constraint.

How can we effectively leverage this structural information?

Bottom-Up: Trust $\hat{y}_{B_1}$ and $\hat{y}_{B_2}$ to build $\hat{y}_{\text{Tot}} = \hat{y}_{B_1} + \hat{y}_{B_2}$

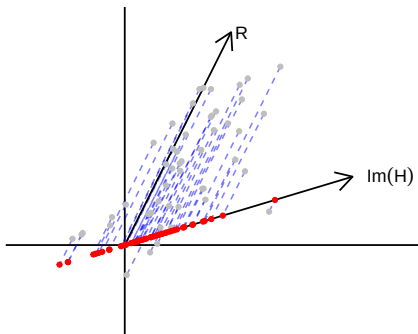
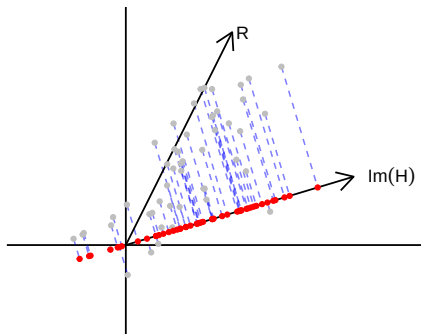
Top-Down: Trust $\hat{y}_{\text{Tot}}$ to build $\tilde{y}_{B_1} = \frac{\hat{y}_{B_1}}{\hat{y}_{B_1} + \hat{y}_{B_2}} \hat{y}_{\text{Tot}}$ and $\tilde{y}_{B_2} = \frac{\hat{y}_{B_2}}{\hat{y}_{B_1} + \hat{y}_{B_2}} \hat{y}_{\text{Tot}}$.

Forecast Reconciliation

Hyndman et al. (2011) consider projection onto the coherent subspace $\text{Im}(H)$:

$$\tilde{\mathbf{y}} = P\hat{\mathbf{y}}.$$

For example,



Forecast Reconciliation: how to choose the projection matrix?

Several methods developed for point forecasting (e.g. [Hyndman et al., 2011](#); [Wickramasuriya et al., 2019](#); [Panagiotelis et al., 2021](#)).

OLS: basic orthogonal projection

$$P_{\text{OLS}} = H(H^{\top}H)^{-1}H^{\top}.$$

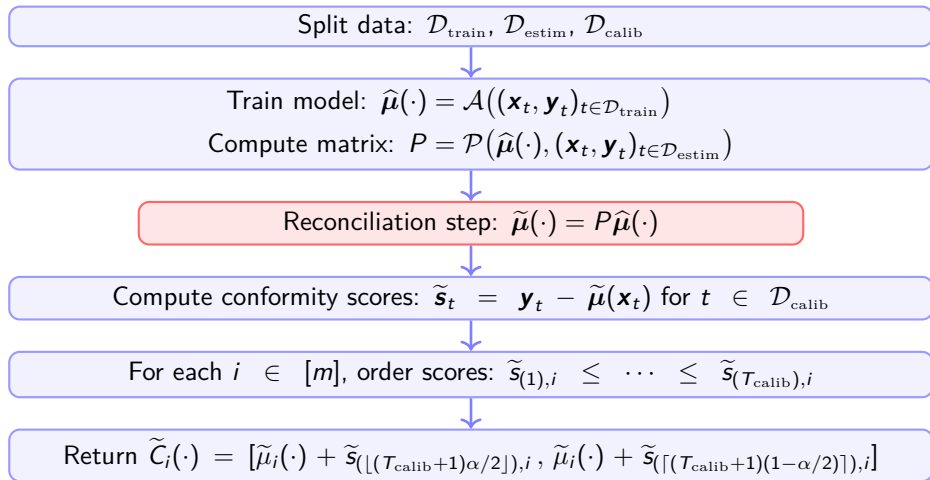
WLS: takes into account the confidence on each individual forecaster

$$P_{\text{WLS}} = H(H^{\top} \text{Diag}(\Sigma)^{-1}H)^{-1}H^{\top} \text{Diag}(\Sigma)^{-1}.$$

MinT: takes into account the confidence AND dependencies within forecasters

$$P_{\text{MinT}} = H(H^{\top}\Sigma^{-1}H)^{-1}H^{\top}\Sigma^{-1}.$$

Reconciled Component-wise SCP



Results (component-wise coverage)

Denote $\widehat{C}(\mathbf{x}_{T+1})$ and $\widetilde{C}(\mathbf{x}_{T+1})$, the conformal regions of component-wise SCP resp. its reconciled version

Under the i.i.d. assumption, for all $i \in [m]$,

$$\mathbb{P}(y_{T+1,i} \in \widetilde{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha \quad \text{and} \quad \mathbb{P}(y_{T+1,i} \in \widehat{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

But no immediate efficiency result (e.g. we don't have $\widetilde{s} \leq \widehat{s}$).

How to define efficiency for component-wise SCP?

Weak version:

$$\text{minimize} \quad \mathbb{E} \left[\sum_{i=1}^m w_i \mathcal{L}_1(C_i(\mathbf{x}_{T+1}))^2 \right],$$

with $\mathbf{w} = (w_1, \dots, w_m)^\top$ a positive weight vector.

Strong version:

$$\forall i \in [m], \quad \text{minimize} \quad \mathbb{E} \left[\mathcal{L}_1(C_i(\mathbf{x}_{T+1}))^2 \right].$$

Additional Assumptions for efficiency results

Disclaimer:

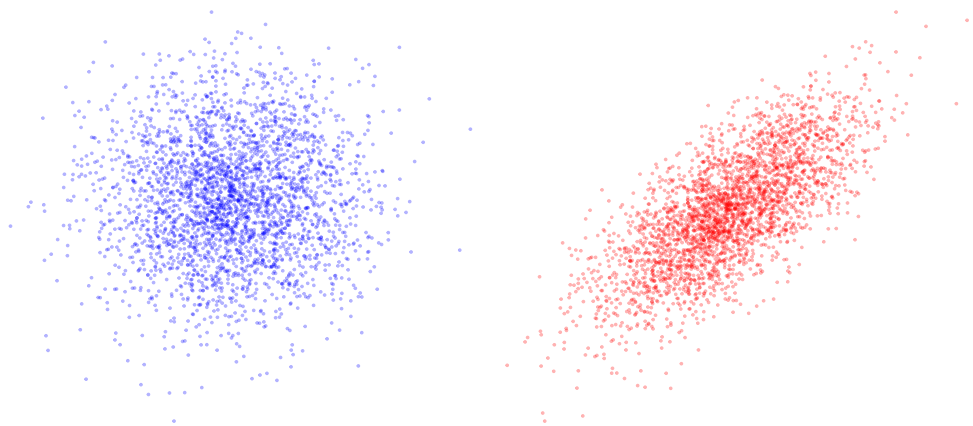
SCP is a **distribution-free** framework that gives **finite-sample** coverage guarantees under mild assumptions.

When it comes to efficiency, additional assumptions are **always** required.

Assumption 1 (weak version): The (i.i.d.) conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ follow some elliptical distribution (parameters are unknown).

Assumption 2 (strong version): The (i.i.d.) conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ follow some elliptical distribution (the variance $\Sigma \succ 0$ exists and is known).

A word on elliptical distribution



Examples: multivariate normal distributions, t -distributions and uniform distribution on an ellipse.

Efficiency results (component-wise SCP)

For any positive vector $\mathbf{w}$, if the reconciled procedure uses $P = H(H^\top \text{diag}(\mathbf{w})H)^{-1}H^\top \text{diag}(\mathbf{w})$ then

Theorem 2: Efficiency – weak version [Principato et al. (2024)]

Under i.i.d. elliptical scores,

$$\mathbb{E} \left[\sum_{i=1}^m w_i \mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\sum_{i=1}^m w_i \mathfrak{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Remark: The practitioner chooses $\mathbf{w}$ to select components for which he does not want to be less efficient.

Efficiency results (component-wise SCP)

If the reconciled procedure uses $P_{\Sigma^{-1}} = H(H^\top \Sigma^{-1} H)^{-1} H^\top \Sigma^{-1}$, then it the same procedure with any other choice of projection matrix P .

Theorem 3: Efficiency – strong version [Principato et al. (2024)]

Under i.i.d. elliptical scores (with known variance $\Sigma \succ 0$),

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

In particular, we also have

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Practical assessment (Component-wise SCP)

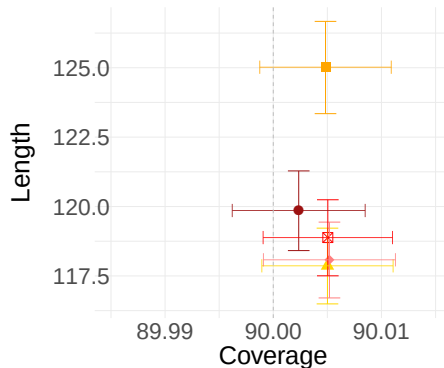
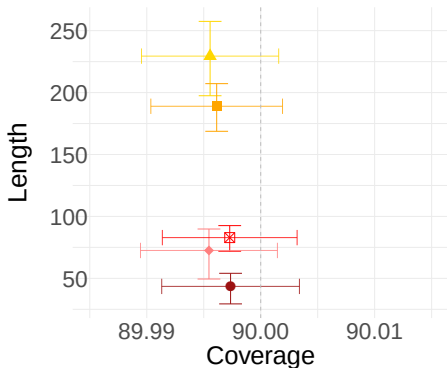
For component-wise coverage, we get valid regions with

Config.	Direct	OLS	WLS	MinT
1	876 $\pm$ 254	787 $\pm$ 226	322 $\pm$ 131	216 $\pm$ 47
2	871 $\pm$ 253	753 $\pm$ 216	308 $\pm$ 116	246 $\pm$ 51
3	3032 $\pm$ 467	2954 $\pm$ 455	1869 $\pm$ 377	1502 $\pm$ 578
4	3036 $\pm$ 479	2901 $\pm$ 458	1581 $\pm$ 340	1404 $\pm$ 571
5	10424 $\pm$ 885	10358 $\pm$ 880	8861 $\pm$ 853	10613 $\pm$ 918
6	10621 $\pm$ 889	10460 $\pm$ 875	7673 $\pm$ 785	10503 $\pm$ 905

Table: root-empirical averages of total squared length $\pm \sqrt{1.96} \times$ standard errors.

Practical assessment (Component-wise SCP)

Methods:  Direct  OLS  WLS  Combi  MinT



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- 4. Gaussian conformal prediction**
 - 4.1 Partially Revealed Outputs
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Extending standardized residuals to multivariate regression

Assume

$$Y|X \sim \mathcal{N}(f(X), \Sigma(X)),$$

where $Y \in \mathbb{R}^k$. We can define the Mahalanobis score ([Braun et al., 2025b](#))

$$S_{\text{Mah}}(X, Y) = \|\Sigma(X)^{-1/2}(Y - f(X))\|_2$$

- Generalization of the standardized residuals ([Lei et al., 2018](#)) in multivariate regression
- Under the Gaussian assumption, $S_{\text{Mah}}(X, Y)|X \sim \chi_2(k)$ which is independent of X

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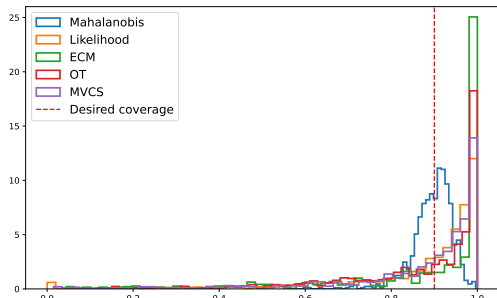
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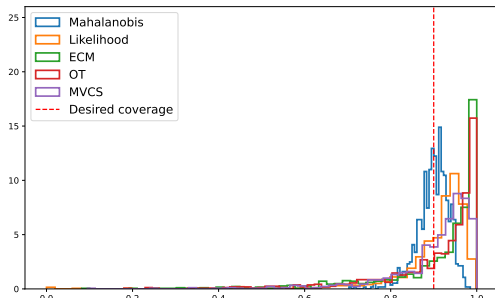
- Generalization of the standardized residuals ([Lei et al., 2018](#)) in multivariate regression
- Under the Gaussian assumption, $S_{\text{Mah}}(X, Y)|X \sim \chi_2(k)$ which is independent of X
- Even if the distribution is not Gaussian, we expect our score to be less dependent on X
 - Let $f(X) = \mathbb{E}[Y|X]$ and $\Sigma(X) = \text{cov}(Y|X)$, and $U = \Sigma(X)^{-1/2}(Y - f(X))$
 - Then $\mathbb{E}[U|X] = 0$ and $\text{cov}(U|X) = I_k$ so the distribution of $S_{\text{Mah}}(X, Y) = \|U\|_2$ should less depend on X
- NB: all developments can be extended to elliptical distributions

Improved Conditional Coverage

- Modeling via ellipsoids $\Rightarrow$ estimation of variance along all directions



Histograms of conditional coverage $\mathbb{P}(Y \in C_\alpha(X_i) \mid X_i)$ on a synthetic dataset ($\alpha = 0.1$)
(left) heteroskedastic Gaussian noise



(right) heteroskedastic exponential noise

- Baselines: empirical covariance matrix ([Johnstone and Cox, 2021](#)), optimal transport ([Thurin et al., 2025](#))

Partially revealed outputs: motivation

- **Example:** predict fasting blood glucose (Y^1) and cholesterol (Y^2)
 - If conformal set too wide, a costly test (e.g., Y^1) may be performed
 - Once Y^1 is revealed, we want to **refine prediction for Y^2**
 - Standard approach: retrain model with $(X, Y^1) \Rightarrow$ expensive

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 - Standard approach: retrain model with $(X, Y^1) \Rightarrow$ expensive
- **Goal: update conformal sets directly** using the probabilistic model
 - Leverage classical conditioning properties of the Gaussian distribution

Method: Gaussian conditioning

- Assuming a Gaussian predictive model:

$$\hat{p}(Y|X) = \mathcal{N}\left(\begin{pmatrix} f^r(X) \\ f^h(X) \end{pmatrix}, \begin{pmatrix} \Sigma^{rr}(X) & \Sigma^{rh}(X) \\ \Sigma^{hr}(X) & \Sigma^{hh}(X) \end{pmatrix}\right)$$

- Partition output into revealed Y^r and hidden Y^h

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- Partition output into revealed Y^r and hidden Y^h
- Conditional distribution (closed form):

$$\hat{p}(Y^h|X, Y^r) = \mathcal{N}(\tilde{f}(X), \tilde{\Sigma}(X))$$

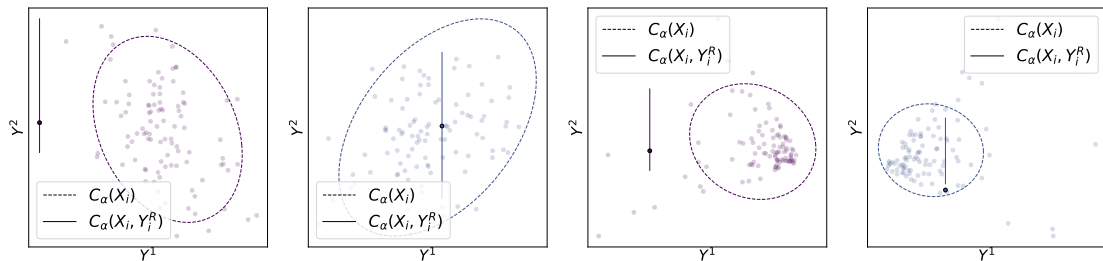
$$\tilde{f}(X) = f^h(X) + \Sigma^{hr}\Sigma^{rr^{-1}}(Y^r - f^r(X)), \quad \tilde{\Sigma}(X) = \Sigma^{hh} - \Sigma^{hr}\Sigma^{rr^{-1}}\Sigma^{rh}$$

- Non-conformity score:

$$S_{\text{Revealed}}(X, Y^r, Y^h) = \|\tilde{\Sigma}(X)^{-1/2}(Y^h - \tilde{f}(X))\|_2$$

Illustration of benefits

- Uses revealed Y^r without retraining
- Produces **narrower, adaptive sets**, and avoids empty sets (robustness)



Left: Gaussian noise

Right: exponential noise.

Different samples from the distribution of $Y|X \sim f(X) + B(X)$
($B(X)$ =heteroskedastic noise and Y^1 is revealed)

Projection of the output: motivation

- High-dimensional predictions: $Y \in \mathbb{R}^k$
- Often (at test time) only a subset or linear combination $\varphi(Y)$ is of interest
- Example:
 - Finance: portfolio returns $(R^1, R^2) = (rY^1, pY^2 + qY^3)$.

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- **Naive approach:** project conformal set $\tilde{C}_\alpha(X)$ onto $\varphi(Y)$
 $\Rightarrow$ conservative (overly large) sets
- **Goal:** construct **direct conformal sets for $\varphi(Y)$** without retraining

Method: linear transformations

- Trained Gaussian model:

$$\hat{p}(Y|X) = \mathcal{N}(f(X), \Sigma(X))$$

- For linear $\varphi(Y) = MY$ ($M \in \mathbb{R}^{p \times k}$):

$$\hat{p}(\varphi(Y)|X) = \mathcal{N}(Mf(X), M\Sigma(X)M^\top)$$

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- Non-conformity score in transformed space:

$$S_{\text{Lin.Trans.}}(X, Y) = \|(M\Sigma(X)M^\top)^{-1/2}(MY - Mf(X))\|_2$$

- Coverage guarantee:

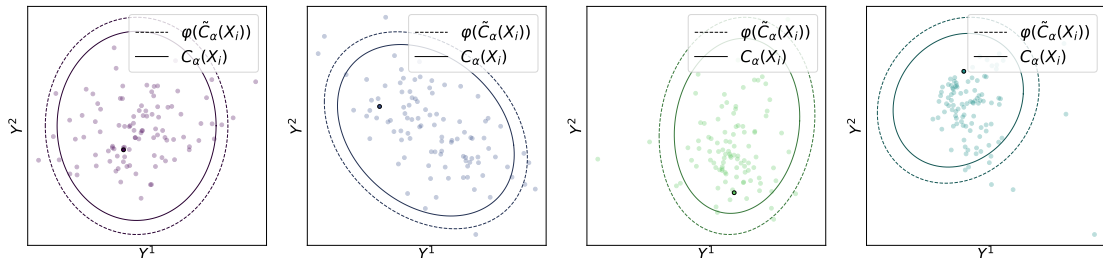
$$\mathbb{P}(MY \in C_\alpha(X)) \in \left[1 - \alpha, 1 - \alpha + \frac{1}{n_2+1}\right)$$

Results: projected vs. direct sets

- Direct conformalization with $S_{\text{Lin.Trans.}}$ (full line) gives:
 - **Tighter sets**, correct coverage (if well-specified)
 - Better adaptation to heteroskedasticity
- Projecting $\tilde{C}_\alpha(X)$ (dashed line) gives **conservative sets**

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Projection of samples from the distribution of $Y|X \sim f(X) + B(X)$
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